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**TITLE: Title:** To visualize temporal data and analyze trends

**Objective:**

To visually represent temporal data in order to identify patterns, trends, and changes over time.

**Expected OUTCOME of Experiment: CO 2**

Detect and understand the stories within datasets and its applications.

**Books/ Journals/ Websites referred:**

1. Data Analysis with Microsoft Excel, Wayne L. Winston, Microsoft Press.
2. Excel 2019 Bible, John Walkenbach, Wiley.
3. Practical Statistics for Data Scientists, Peter Bruce and Andrew Bruce, O'Reilly Media

**Background Theory:**

This experiment focuses on the visualization of temporal data to facilitate effective analysis of patterns and trends over time. Temporal data, which is indexed by time, is commonly encountered in domains such as data analytics, economics, weather forecasting, and system monitoring. By representing such data using appropriate visualization techniques like line charts, time-series plots, and trend lines, complex variations and long-term behaviors become easier to interpret. Visual analysis helps in identifying seasonal effects, growth or decline trends, anomalies, and cyclic patterns, thereby supporting informed decision-making and predictive analysis.

**Dataset link:** Household Electricity Consumption Records

<https://www.kaggle.com/datasets/thedevastator/240000-household-electricity-consumption-records>

## Colab Workbook:

<https://colab.research.google.com/drive/1RIoyeJfHrMaqqU9O-3QOA6bIeZAtpOF?usp=sharing>

## Conclusion:

1. The **heatmap** shows average power usage across different hours and days. Power consumption is lowest during early morning hours (2 AM–5 AM) and increases during the day. The highest usage occurs in the evening (6 PM–9 PM), indicating peak household activity.
2. The **bar graph** represents hourly average power consumption. It shows a gradual increase from morning, peak consumption in the evening, and a decline at night. This reflects normal residential electricity usage patterns.
3. The **treemap** shows the percentage contribution of each sub-meter:
  - Sub\_metering\_3 contributes the most energy.
  - Sub\_metering\_2 contributes moderately.
  - Sub\_metering\_1 contributes the least.

This indicates that heating/cooling and major appliances consume the highest electricity.

4. The **monthly graph** shows fluctuations in energy consumption. There is no strong upward or downward trend, but variations suggest seasonal influence.
5. The **moving average** smooths short-term fluctuations and shows that overall consumption remains relatively stable over time.

## Post Laboratory Questions:

Following questions are to be answered in this document.

### 1. How do moving averages help in identifying trends?

- Moving averages smooth out price trends by filtering out the noise from random short-term price fluctuations.
- By calculating the moving average, the impacts of random, short-term fluctuations on the price of a stock over a specified time frame are mitigated.
- A rising moving average indicates that the security is in an uptrend, while a declining moving average indicates that it is in a downtrend.
- Moving averages can help investors gauge market trends by smoothing out price volatility and showing the market's relative direction.

### 2. How can you detect anomalies in a time-series dataset?

Anomalies can be detected by:

- Identifying sudden spikes or drops in graphs.
- Comparing actual values with moving averages.
- Using statistical methods like standard deviation or z-scores.
- Applying anomaly detection models in Python.

### 3. How do you analyze seasonal patterns using Python?

Seasonal patterns can be analyzed by:

- Grouping data by month, day, or hour using `groupby()`.
- Using `resample()` to aggregate data.
- Plotting seasonal decomposition using `seasonal_decompose()` from `statsmodels`.
- Visualizing patterns with line plots or heatmaps.

### 4. Explain how drilling down into specific months or years helps in understanding the data

Drilling down allows detailed analysis of specific periods.  
It helps identify:

- Seasonal effects
  - Monthly variations
  - Sudden changes in specific years
- This improves understanding of trends and irregular patterns.

5. What patterns can you observe in the airline passenger dataset from 1949–1960?

The airline passenger dataset shows:

- A clear upward trend in passenger numbers over the years.
- Strong seasonal patterns (peak travel during mid-year months).
- Increasing seasonal variation over time (both trend and seasonality grow).

This indicates consistent growth in air travel demand between 1949 and 1960.